

Homework 6

Due: May. 25th

Turn in your homework in class

Rules:

- Please work on your own. Discussion is permissible, but extremely similar submissions will be judged as plagiarism!
- Please show all intermediate steps: a correct solution without an explanation will get zero credit.
- Please submit **on time**. No late submission will be accepted.
- Please prepare your submission in English only. No Chinese submission will be accepted.
- Please retain **3** decimal places if rounding is required without special requirements for the question,

1

[10 points] For the circuit in **Figure 1**, calculate the following variables for the power absorbed by the load:

- (a) the power factor,
- (b) the real power,
- (c) the reactive power,
- (d) the apparent power,
- (e) the complex power.

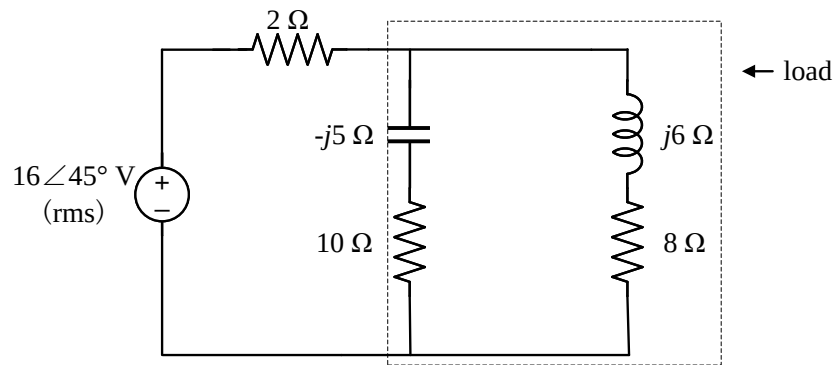


Figure 1

2

[10 points] The resistor R in **Figure 2** equals to $5\ \Omega$. Calculate the real power and the reactive power delivered to R .

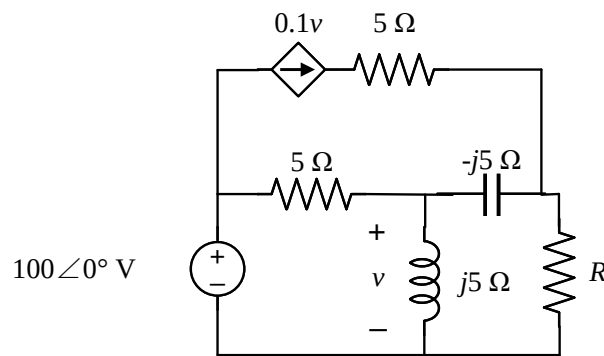


Figure 2

3

[16 points] **Figure 3** shows a sinusoidal steady-state circuit with coupled coils. Take $\omega = 1000$ rad/s.

- (a) Determine the phasor currents $\mathbf{I}_1$, $\mathbf{I}_2$ and $\mathbf{I}_3$ in the circuit.
- (b) Find the energy stored in the coupled coils at $t = 2$ ms.

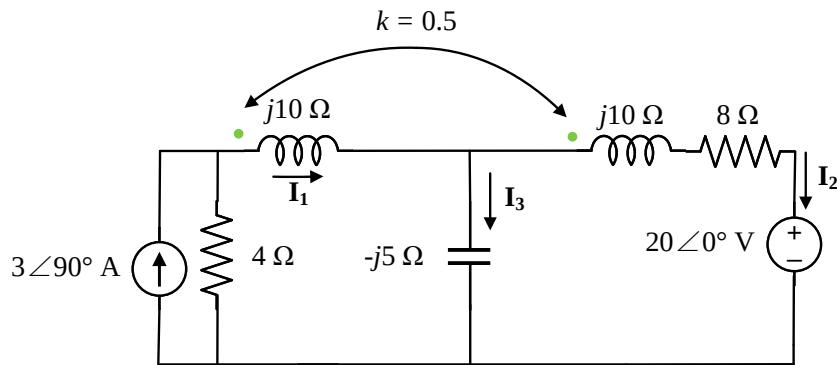


Figure 3

4

[16 points] Find $\mathbf{I}_1$ and $\mathbf{I}_2$ in the ideal transformer circuit of **Figure 4**.

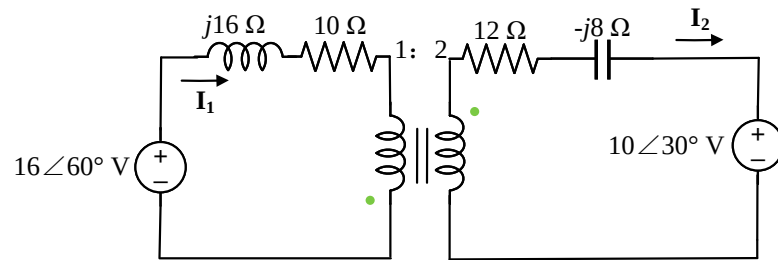


Figure 4

5

[16 points] For the ideal transformer network in **Figure 5**, find:

- (a) the complex power supplied by the source,
- (b) the average power delivered to the $18\ \Omega$ resistor.

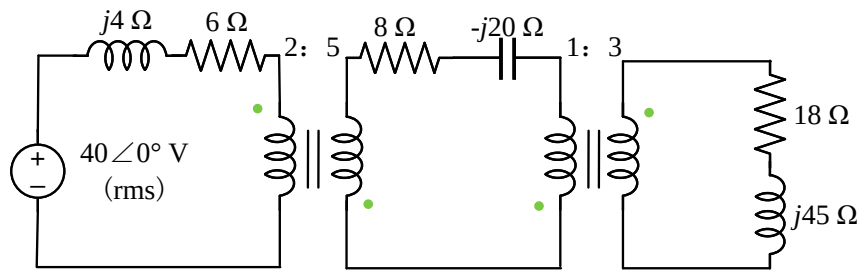


Figure 5

6

[16 points] A balanced three-phase $Y - \Delta$ system has $\mathbf{V}_{an} = 220 \angle 0^\circ$ V rms and $\mathbf{Z}_\Delta = 51 + j45 \Omega$. If the line impedance per phase is $0.4 + j1.2 \Omega$, find the line voltage, line current and the total complex power delivered to the load.

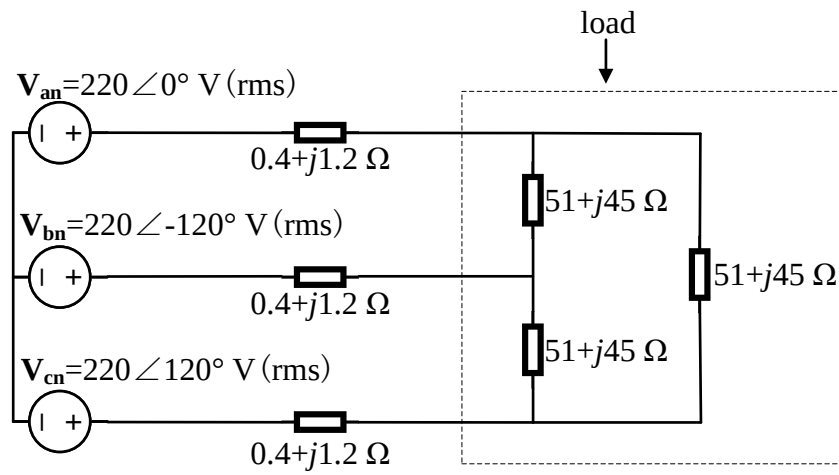


Figure 6

7

[16 points] Find the maximum average power absorbed by the load in the circuit of **Figure 7** when:

- (a) The load is a pure resistor,
- (b) The load is an impedance.

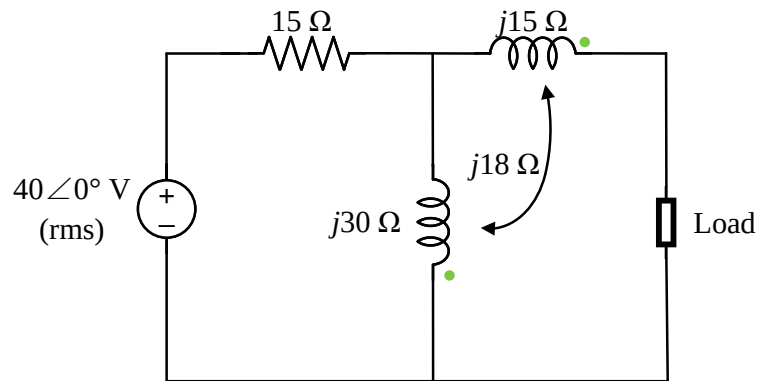


Figure 7